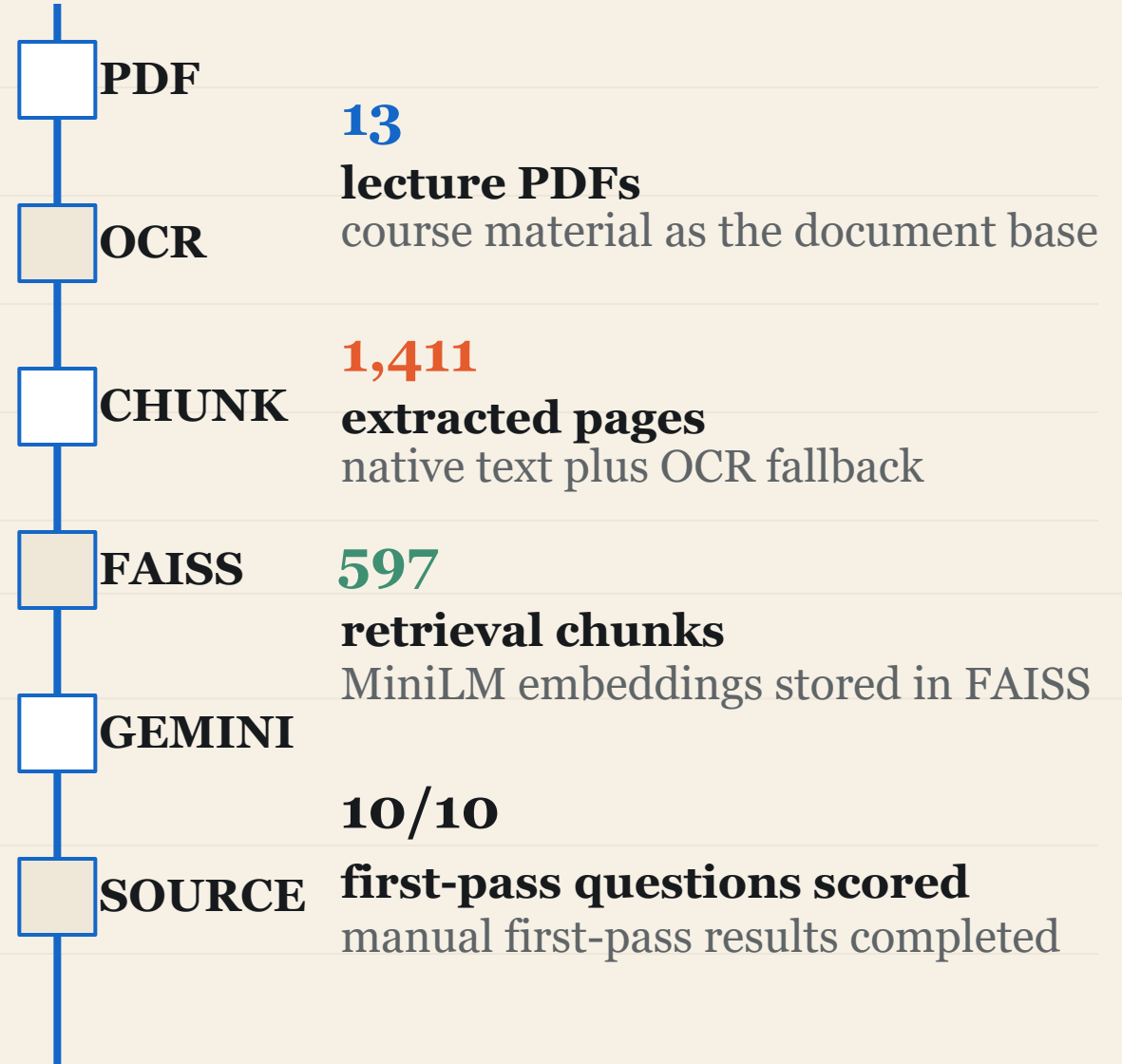


RAG-Based Course Assistant

A lecture-grounded assistant for Generative AI course questions, evaluated against a plain Gemini baseline.



Xiaohang Cheng(xc2856), Shuzhi Yang(sy3321), Rongrong Chen(rc3814)

The central result: RAG makes course-grounding visible, even when it does not always beat Gemini on correctness

What was built, what was compared, and what we learned from the first-pass evaluation.

Built

13 PDFs ->
597 chunks ->
FAISS index

Compared

Plain Gemini
vs grounded
RAG answers

Learned

Retrieval
relevance is
the bottleneck

PROJECT THESIS

A strong baseline can answer common definitions. The RAG system matters because it exposes the lecture evidence and makes retrieval failures diagnosable.

Demo: ask one course question, show retrieved chunks, generate an answer, then inspect source pages.

A generic LLM can answer many course questions, but it cannot show where the course said it

The project targets students who need answers grounded in lecture slides, not broad textbook-style explanations.

Plain Gemini baseline

Answers directly from model priors. It can be correct on common definitions, but it does not inspect course material or expose source pages.

Risk

Correct-looking answers may drift away from the instructor's lecture framing.



RAG course assistant

Retrieves lecture chunks first, injects them into a grounded prompt, and asks Gemini to answer only from those chunks.

Goal

Make answers auditable: users can inspect source lecture pages behind the response.

The research question is narrower than "can we build a chatbot?"

We test whether course-specific retrieval changes answer grounding, and where retrieval breaks.

Can a RAG system produce course answers that are more grounded in lecture material than a plain LLM baseline?

Hypothesis: retrieval may not always improve broad correctness, but it should improve source visibility and faithfulness when the right chunks are retrieved.

Scope

10 course-definition/concept questions, all completed in the final evaluation after routing Gemini API calls through Agicto.

Assumption

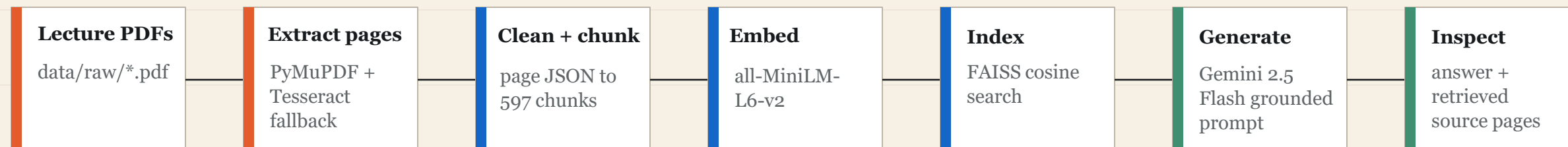
Course material is the authority; retrieved chunks are the allowed context for RAG answers.

Boundary

No full production search stack, no user study, no claim that RAG beats Gemini on every accuracy metric.

The system is a reproducible retrieval pipeline from lecture PDF to cited answer

Each stage creates an inspectable artifact, which helps debugging and grading.



Generated artifacts

all_pages.json | all_chunks.json | faiss.index | chunk_metadata.json | results_first_pass.json

Design choice: keep the answer model constrained by retrieved context so failures become visible instead of silently hidden.

The contribution is a compact RAG lab with baseline comparison, not a one-off demo prompt

Rubric alignment: architecture, integration strategy, performance/reproducibility, and practical impact.

OCR-aware ingestion

Native PDF extraction falls back to Tesseract for low-text pages, preserving page-level provenance.

Retrieval-ready corpus

Light cleanup, page-aware chunks, normalized embeddings, and FAISS index artifacts.

Grounded generation contract

the answer model is constrained to retrieved course material

Evaluation harness

Baseline vs RAG scripts save outputs, retry free-tier rate limits, and enable manual scoring.

Practical impact: students can ask natural course questions and inspect the lecture evidence behind an answer.

The live demo should show retrieval before generation, so the audience can judge grounding.

A single question can demonstrate query, retrieved chunks, answer, sources, and optional baseline comparison.

What is the difference between semantic search and RAG?

Retrieved Sources

[1] lecture_04.pdf pages 4-4 | score=0.7437
[2] lecture_04.pdf pages 3-3 | score=0.5166

RAG Answer

Semantic search retrieves relevant documents by meaning; RAG retrieves relevant information and uses it to generate a contextual response.
Sources: lecture_04.pdf, pages 3-4.



The evaluation separates broad correctness from course-grounding and retrieval relevance

This is important because Gemini can be correct even when it is not grounded in course slides.

Metric	What it tests	Why it matters
baseline_correctness	Gemini answer quality without retrieval	baseline comparison
rag_correctness	RAG final answer correctness	end-user answer quality
rag_groundedness	faithfulness to retrieved chunks	course-material discipline
retrieval_relevance	whether top chunks match the question	root-cause diagnosis

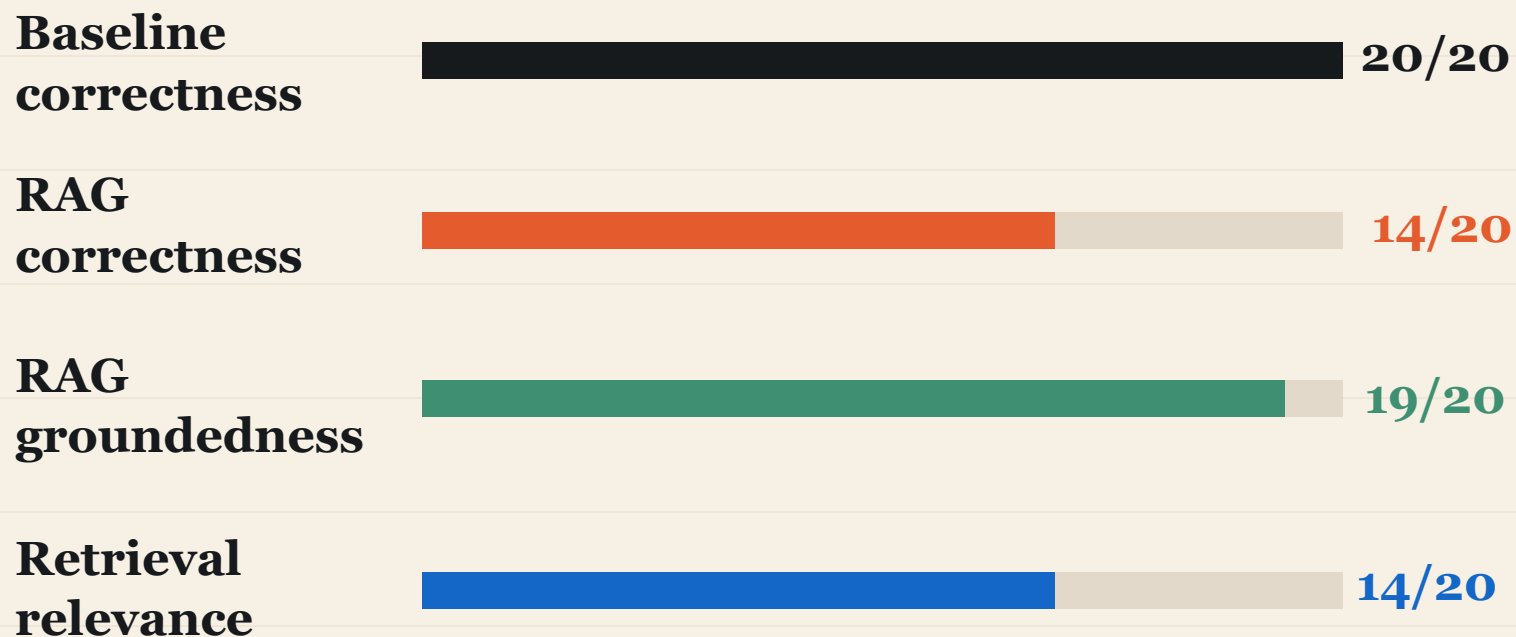
10
planned questions
q1-q10 course
topics

10
completed first pass
results saved in eval/

0-2
manual score scale
correctness, grounding,
relevance

RAG is highly grounded when it answers, but retrieval relevance is the limiting metric

First-pass manual scores show strong source discipline and clear failure cases.



Pattern

Strong cases: q1 RAG, q2 RLHF, q3 bias-variance, q5 semantic search vs RAG, q6 benchmarking, q9 batching.

Failure cases: q4 LoRA and q8 consumer hardware, q10 prompt tuning vs LoRA.

Manual scores across q1-q10; each metric max = 2 points per question.

Retrieval ablation: top-k sweep

A small top-k ablation shows that moderate retrieval depth works best overall. Top-k = 4 gave the best overall balance between useful context and retrieval noise.

Question	k=2	k=4	k=6	Best
What is RAG?	Good	Strong	Slightly noisy	4
What is RLHF?	Good	Strong	Similar but nosier	4
What is the bias-variance tradeoff?	Incomplete	Strong	Strong but verbose	4
what is LoRA?	Fail	Fail	Improves	6
Semantic search vs RAG	Strong	Strong	Similar	4

k=2 is often too narrow and can miss necessary supporting context.
k=6 can help harder queries like LoRA, but it also introduces more irrelevant chunks.
k=4 is the best default for the final demo and evaluation.

The main bottleneck is retrieval quality; OCR/text noise is a secondary issue

The RAG prompt behaved correctly by admitting insufficient context when retrieval failed.

q4 / LoRA

retrieved SLoRA serving and rank slides, but missed a definition chunk

retrieval=0/2

q8 / consumer hardware

retrieved serving-cost and batching material instead of training constraints

retrieval=0/2

Next
ablations

hybrid BM25 +
dense search

query expansion
for acronyms

reranker over
top-k chunks

OCR cleanup for
mojibake

top-k and chunk-
size sweep

The repo tells a reproducible story: build the corpus, index it, compare systems, demo it

This closes the GitHub and demo portions of the rubric while staying honest about remaining experiments.

Reproducible run path

```
$ python src/extract_pdf.py
$ python src/chunk_text.py
$ python src/build_index.py
$ python src/evaluate_systems.py
$ streamlit run app/streamlit_app.py
```

What is done

End-to-end RAG pipeline, Streamlit interface, baseline comparison, manual first-pass scoring.

What remains

- strengthen retrieval for LoRA and consumer-hardware failure cases
- test hybrid retrieval or reranking
- improve OCR cleanup for visual slides
- harden low-confidence warning behavior in the demo

Final thesis: RAG does not automatically beat a strong baseline; its value is making course-grounding visible, testable, and improvable.

Thank you for your patience!